CHAPTER 3

Determinants

行列式

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Chapter 3 Determinants

- § 3.1 Introduction to Determinants
- § 3.2 Properties of Determinants
- § 3.3 Cramer's Rule, Volume, and Linear Transformations

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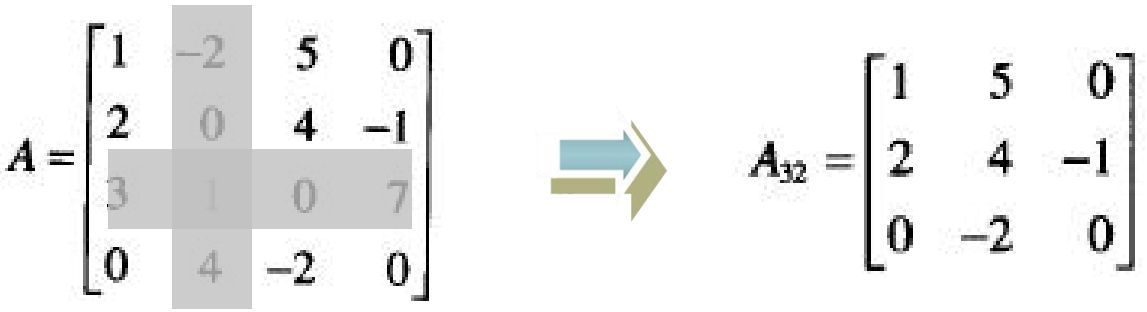
3.1 Introduction to Determinants

A determinant is a **number** that is assigned to a **square** array of numbers in a **certain** way.

- In Cauchy's day, when life was simple and matrices were small, determinants played major role in analytic geometry and other parts of mathematics.
- Today, determinants are of little numerical value in large-scale matrix computations that occur so often. Nevertheless, determinantal formulas still give important information about matrices.

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- For any square matrix A, let A_{ij} denote the submatrix formed by deleting the ith row and jth column of A. For instance,



- We can now give a recursive definition of a determinant. When n = 3, det A is defined using determinants of the 2 x 2 submatrices A_{1j}, When n = 4, det A uses determinants of the 3 x 3 submatrices A_{1j}. In general, n x n determinants is defined by determinants of (n-1)x(n-1) submatrices.

§ 3.1 Introduction to Determinants

- **Definition：** For $n \geq 2$, the determinant of an $n \times n$ matrix $A = [a_{ij}]$ is the sum of n terms of the form $\pm a_{ij} \det A_{ij}$, with plus and minus signs alternating, where the entries $a_{11}, a_{12}, \ldots, a_{1n}$ are from the first row of A. In symbols,

$$\det A = a_{11} \cdot \det A_{11} - a_{12} \cdot \det A_{12} + \cdots + (-1)^{1+n} a_{1n} \cdot \det A_{1n}$$

$$= \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det A_{1j}$$

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- Example：Computer the determinant of

$$A=\begin{bmatrix}1 & 5 & 0\\ 2 & 4 & -1\\ 0 & -2 & 0\end{bmatrix}$$

- Solution：Compute

$$\begin{aligned}\det A &= a_{11}\cdot\det A_{11}-a_{12}\cdot\det A_{12}+a_{13}\cdot\det A_{13}.\\ &=1\cdot\det\begin{bmatrix}4 & -1\\ -2 & 0\end{bmatrix}-5\cdot\det\begin{bmatrix}2 & -1\\ 0 & 0\end{bmatrix}+0\cdot\det\begin{bmatrix}2 & 4\\ 0 & -2\end{bmatrix}\\ &=1(0-2)-5(0-0)+0(-4-0)=-2\end{aligned}$$

- Example：Compute the determinant of

Sol.

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ 2 & 0 & 1 \end{bmatrix}$$

$$\det A = 1\det\begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix} - 2\det\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} + 0\det\begin{bmatrix} 3 & -1 \\ 2 & 0 \end{bmatrix}$$

So

$$\begin{vmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ 2 & 0 & 1 \end{vmatrix} = 1\begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} - 2\begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} + 0\begin{vmatrix} 3 & -1 \\ 2 & 0 \end{vmatrix} = 1$$

§ 3.1 Introduction to Determinants

- Given A = [a_{ij}], the (*i*, *j*)-cofactor(余因子) of A is the number C_{ij} given by

$$C_{ij} = (-1)^{i+j} \det A_{ij} \qquad (4)$$

- Then

$$\det A = a_{11} \cdot C_{11} + a_{12} \cdot C_{12} + \cdots + a_{1n} \cdot C_{1n}$$

This formula is called a cofactor expansion(余因子展开式) across the first row of A.

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§ 3.1 Introduction to Determinants

Theorem 1

The determinant of an n × n matrix A can be computed by a cofactor expansion across any row or down any column. The expansion across the ith row using the cofactors in (4) is

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

The cofactor expansion down the jth column is

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

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- Example ： Use a cofactor expansion across the third row to compute det *A*, where

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

- Solution： Compute

$$\begin{aligned} \det A &= a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33} \\ &= (-1)^{3+1} a_{31} \det A_{31} + (-1)^{3+2} a_{32} \det A_{32} + (-1)^{3+3} a_{33} \det A_{33} \\ &= 0\begin{vmatrix} 5 & 0 \\ 4 & -1 \end{vmatrix} - (-2)\begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix} + 0\begin{vmatrix} 1 & 5 \\ 2 & 4 \end{vmatrix} = 0 + 2(-1) + 0 = -2 \end{aligned}$$

- Example ： Compute det A, where

$$A=\begin{bmatrix}3 & -7 & 8 & 9 & -6\\ 0 & 2 & -5 & 7 & 3\\ 0 & 0 & 1 & 5 & 0\\ 0 & 0 & 2 & 4 & -1\\ 0 & 0 & 0 & -2 & 0\end{bmatrix}$$

- Solution： Compute

$$\det A = 3\cdot\begin{bmatrix}2 & -5 & 7 & 3\\ 0 & 1 & 5 & 0\\ 0 & 2 & 4 & -1\\ 0 & 0 & -2 & 0\end{bmatrix} - 0\cdot C_{21} + 0\cdot C_{31} - 0\cdot C_{41} + 0\cdot C_{51}$$

$$= 3\cdot 2\cdot\begin{bmatrix}1 & 5 & 0\\ 2 & 4 & -1\\ 0 & -2 & 0\end{bmatrix} = 3\cdot 2\cdot(-2) = -12$$

Example：

Compute the determinant of $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 1 & 5 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 3 & 5 \end{bmatrix}$

Sol：

$$\begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 1 & 5 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 3 & 5 \end{vmatrix} = 1 \begin{vmatrix} 2 & 1 & 5 \\ 0 & 2 & 1 \\ 0 & 3 & 5 \end{vmatrix} - 0 \begin{vmatrix} 2 & 3 & 4 \\ 0 & 2 & 1 \\ 0 & 3 & 5 \end{vmatrix} + 0 \begin{vmatrix} 2 & 3 & 4 \\ 2 & 1 & 5 \\ 0 & 3 & 5 \end{vmatrix} - 0 \begin{vmatrix} 2 & 3 & 4 \\ 2 & 1 & 5 \\ 0 & 2 & 1 \end{vmatrix}$$

$$= 1 \cdot 2 \begin{vmatrix} 2 & 1 \\ 3 & 5 \end{vmatrix} = 14$$

§ 3.1 Introduction to Determinants

Theorem 2

If A is a triangular matrix, then det *A* is the product of the entries on the main diagonal of *A*.

Triangular Matrices:

$$\begin{bmatrix} * & * & \cdots & * & * \\ 0 & * & \cdots & * & * \\ 0 & 0 & \ddots & * & * \\ 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 0 & * \end{bmatrix} \qquad \begin{bmatrix} * & 0 & 0 & 0 & 0 \\ * & * & 0 & 0 & 0 \\ * & * & \ddots & 0 & 0 \\ * & * & \cdots & * & 0 \\ * & * & \cdots & * & * \end{bmatrix}$$

(upper triangular) (lower triangular)

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§ 3.1 Introduction to Determinants

- Example：

$$\begin{vmatrix} 2 & 3 & 4 & 5 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & -3 & 5 \\ 0 & 0 & 0 & 4 \end{vmatrix} = 2*1*(-3)*4= -24$$

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Assignment

- Ex3.1： 33-36

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§ 3.2 Properties of Determinants

- 1. Properties of Determinants
- 2. Column Operation
- 3. Determinants and Matrix Products

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§ 3.2 Properties of Determinants

Properties of Determinants

Theorem 3 Row Operations

Let A be square matrix.

a. If a multiple of one row of A is added to another row to produce a matrix B, then $\det B = \det A$.

b. If two rows of A are interchanged to produce B, then $\det B = -\det A$.

c. If one row of A is multiplied by k to produce B, then $\det B = k\det A$.

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Proof：Theorem 3 Can be reformulated as follows:

If A is an n*n matrix and E is an n*n elementary matrix. Then det EA = (det E)(detA) where

$$\det E = \begin{cases} 1 & \text{若}E\text{是一个行倍加} \\ -1 & \text{若}E\text{是一个交换} \\ r & \text{若}E\text{是一个}r\text{倍乘} \end{cases}$$

The proof is by induction on the size of A.

Basis: The case of a 2*2 matrix was verified in Exercises 33-36 of Section 3.1.

Induction: Suppose the theorem has been verified for determinants of k*k matrices with k ≥2.

Let n = k+1, and let A be n*n.

The action of E on A involves either two rows or only one row.

We can expand *det EA* across a row that is unchanged by the action *E*, say, row i.

Let A_{ij} (respectively B_{ij}) be the matrix obtain by deleting row i and column j from A (respectively B=EA).

Then the row of B_{ij} are obtained form the rows of A_{ij} by the same type of elementary row operation that E performs on A.

Since these submatrices are only k*k, the induction assumption implies that $\det B_{ij} = \alpha \cdot \det A_{ij}$

where $\alpha = 1, -1, r$, depending on the nature of E.

Thus, the cofactor expansion of EA across row i is

$$\begin{aligned}\det EA &= a_{i1}(-1)^{i+1} \det B_{i1} + \cdots + a_{in}(-1)^{i+n} \det B_{in} \\ &= \alpha\, a_{i1}(-1)^{i+1} \det A_{i1} + \cdots + \alpha\, a_{in}(-1)^{i+n} \det A_{in} \\ &= \alpha \cdot \det A\end{aligned}$$

- Example： Compute det A, where

$$A=\begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}.$$

- Solution： Compute

$$\det A=\begin{vmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{vmatrix}=\begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ -1 & 7 & 0 \end{vmatrix}=\begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 2 \end{vmatrix}$$

An interchange of rows 2 and 3 reverses the sign of the determinant, so

$$\det A=-\begin{vmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{vmatrix}=-(1)(3)(-5)=15$$

- Example ： Compute det A, where

$$A=\begin{bmatrix} 0 & 1 & 2 & -1 \\ 2 & 5 & -7 & 3 \\ 0 & 3 & 6 & 2 \\ -2 & -5 & 4 & -2 \end{bmatrix}$$

- Solution ： Compute

$$\det A=\begin{vmatrix} 0 & 1 & 2 & -1 \\ 2 & 5 & -7 & 3 \\ 0 & 3 & 6 & 2 \\ 0 & 0 & -3 & 1 \end{vmatrix}=-2\begin{vmatrix} 1 & 2 & -1 \\ 3 & 6 & 2 \\ 0 & -3 & 1 \end{vmatrix}=-2\begin{vmatrix} 1 & 2 & -1 \\ 0 & 0 & 5 \\ 0 & -3 & 1 \end{vmatrix}$$

$$=(-2)\cdot(1)\begin{bmatrix} 0 & 5 \\ -3 & 1 \end{bmatrix}=-2\cdot(15)=-30$$

§ 3.2 Properties of Determinants

- A common use of Theorem 3 is to factor out a common multiple of one row of a matrix.

c. If one row of A is multiplied by k to produce *B*, then det *B* = *k*det *A*.

$$\begin{vmatrix} * & * & * \\ 5k & -2k & 3k \\ * & * & * \end{vmatrix} = k \begin{vmatrix} * & * & * \\ 5 & -2 & 3 \\ * & * & * \end{vmatrix}$$

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Example:

$$\text{Compute} \begin{vmatrix} 2 & 4 & 6 \\ 5 & 6 & 7 \\ 7 & 6 & 10 \end{vmatrix}$$

Sol.

$$\begin{vmatrix} 2 & 4 & 6 \\ 5 & 6 & 7 \\ 7 & 6 & 10 \end{vmatrix} = 2 \begin{vmatrix} 1 & 2 & 3 \\ 5 & 6 & 7 \\ 7 & 6 & 10 \end{vmatrix} = 2 \begin{vmatrix} 1 & 2 & 3 \\ 0 & -4 & -8 \\ 0 & -8 & -11 \end{vmatrix}$$

$$= 2(-4) \begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -8 & -11 \end{vmatrix} = 2(-4) \begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 5 \end{vmatrix} = 2(-4)(1)(1)(5) = -40$$

§ 3.2 Properties of Determinants

- Suppose a square matrix A has been reduced to an echelon form U by row replacements and row interchanges.(This is always possible.)
- If there are *r* interchanges, then Theorem 3 shows that

$$\det A = (-1)^r \det U$$

- Thus

$$\det A = \begin{cases} = 0, & \text{当} A \text{不可逆} \\ \neq 0, & \text{当} A \text{可逆} \end{cases}$$

$$U = \begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & \blacksquare \end{bmatrix} \quad \det U \neq 0$$

$$U = \begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & \blacksquare \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \det U = 0$$

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§ 3.2 Properties of Determinants

Theorem 4

A square matrix A is invertible if and only if det A ≠ 0.

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§ 3.2 Properties of Determinants

Theorem 5

If A is an $n \times n$ matrix, then $\det A^T = \det A$.

The above result implies that each statement in Theorem 3 is true when the word "row" is replaced by "column".

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- Proof:

when n = 1, the theorem is obvious.

Suppose the theorem is true for k*k determinants and let n = k+1.

The cofactor of a_{1j} in A equals the cofactor of a_{j1} in A^T.

Hence the cofactor expansion of det A along the first row equals the cofactor expansion of det A^T down the first column.

That is A and A^T have equal determinants.

By the principle of induction, the theorem is true for all n≥1.

§ 3.2 Properties of Determinants

3. Determinants and Matrix Products

Theorem 6

If A and B are $n \times n$ matrices, then

$$\det AB = (\det A)(\det B).$$

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Proof:

If A is not invertible, the result is obvious.

If A is invertible, then A and the identity matrix I are row equivalent by the Invertible Matrix Theorem.

So there exist elementary matrices $E_1, \cdots, E_p$ such that

$$A = E_p E_{p-1} \cdots E_1 I_n = E_p E_{p-1} \cdots E_1$$

For brevity, write $|A|$ for det A. Then repeated application of Theorem 3, as rephrased above, show that

$$\begin{aligned} |AB| &= |E_p \cdots E_1 B| = |E_p| \cdot |E_{p-1} \cdots E_1 B| = \cdots \\ &= |E_p| \cdots |E_1| \cdot |B| = \cdots = |E_p \cdots E_1| \cdot |B| \\ &= |A| \ |B| \end{aligned}$$

§ 3.2 Properties of Determinants

- Example : Verify Theorem 6 for $A=\begin{bmatrix}6 & 1\\ 3 & 2\end{bmatrix}$, $B=\begin{bmatrix}4 & 3\\ 1 & 2\end{bmatrix}$

Solution: $\because$ $AB=\begin{bmatrix}6 & 1\\ 3 & 2\end{bmatrix}\begin{bmatrix}4 & 3\\ 1 & 2\end{bmatrix}=\begin{bmatrix}25 & 20\\ 14 & 13\end{bmatrix}$

$$\det AB = 25\cdot 13 - 20\cdot 14 = 325 - 280 = 45$$

$$\det A = 9\ ,\quad \det B = 5\ ,$$

$$\therefore\quad (\det A)(\det B) = 9\cdot 5 = 45 = \det AB$$

- warning: det(A + B) != detA + detB

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Assignment

- Ex3.2: 5,9,31,35

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§ 3.3 Cramer's Rule, Volume, and Linear Transformations

- 1. Cramer's Rule
- 2. A Formula for A^{-1}
- 3. Determinants as Area or Volume
- 4. Linear Transformations

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§ 3.3 Cramer's Rule, Volume, and Linear Transformations

Cramer's Rule (克莱姆法则)

- Cramer's rule is needed in a variety of theoretical calculations.
- It can be used to study how the solution of Ax = b is affected by changes in the entries of b.
- For any n x n matrix A and b in R^n, let $A_i(b)$ be the matrix obtained from A by replacing column i by the vector b.

$$A_i(\boldsymbol{b}) = [\boldsymbol{a}_1 \ \cdots \ \boldsymbol{b} \ \cdots \ \boldsymbol{a}_n]$$

第 i 列

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§ 3.3 Cramer's Rule, Volume, and Linear Transformations

Cramer's Rule

Theorem 7 Cramer's Rule

Let A be an invertible n×n matrix. For any b in R^n, the unique solution x of Ax = b has entries given by

$$x_i = \frac{\det A_i(b)}{\det A}, \quad i = 1, 2, \cdots, n \qquad (1)$$

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- Proof ：

Denote the columns of A by a_1 ,…, a_n and the columns of the n*n identity matrix I by e_1,…,e_n. If Ax = b, the definition of matrix multiplication shows that

$$A \cdot I_i(\boldsymbol{x}) = A[\boldsymbol{e}_1 \ \cdots \ \boldsymbol{x} \ \cdots \boldsymbol{e}_n] = [A\boldsymbol{e}_1 \ \cdots \ A\boldsymbol{x} \ \cdots \ A\boldsymbol{e}_n]$$
$$= [\boldsymbol{a}_1 \cdots \ \boldsymbol{b} \ \cdots \boldsymbol{a}_n] = A_i(\boldsymbol{b})$$

By the multiplicative property of determinants,

$$(\det A)(\det I_i(\boldsymbol{x})) = \det A_i(\boldsymbol{b})$$

The second determinant on the left is simply x_i.

$$(\det A)\boldsymbol{x}_i = \det A_i(\boldsymbol{b}) .$$

Hence, This proves (1) because A is invertible And det A ≠0

- Example ： Use Cramer's rule to solve the system

$$3x_1 - 2x_2 = 6$$
$$-5x_1 + 4x_2 = 8$$

Solution : View the system as A**x** = **b**.

$$A = \begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix}, \quad A_1(\boldsymbol{b}) = \begin{bmatrix} 6 & -2 \\ 8 & 4 \end{bmatrix}, \quad A_2(\boldsymbol{b}) = \begin{bmatrix} 3 & 6 \\ -5 & 8 \end{bmatrix}$$

Since det A = 2, the system has a unique solution. By Cramer's rule,

$$x_1 = \frac{\det A_1(\boldsymbol{b})}{\det A} = \frac{24+16}{2} = 20$$
$$x_2 = \frac{\det A_2(\boldsymbol{b})}{\det A} = \frac{24+30}{2} = 27$$

A Formula for A^{-1}

The jth column of A^{-1} is a vector $\mathbf{x}$ that satisfies $Ax = e_j$

By Cramer's rule, $\{A^{-1}\text{中}(i,j)\text{元素}\} = x_i = \dfrac{\det A_i(e_j)}{\det A}$

A cofactor expansion down column i of $A_i(e_j)$ shows that

$$\det A_i(e_j) = (-1)^{i+j} \det A_{ji} = C_{ji}$$

Thus

$$A^{-1} = \frac{1}{\det A}\begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

The matrix of cofactors is called the **adjugate (伴随矩阵)** of A, denoted by **adj A.**

§ 3.3 Cramer's Rule, Volume, and Linear Transformations

Theorem 8 An Inverse Formula

Let A be an invertible n × n matrix. Then

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A$$

This theorem is useful for theoretical calculations, but not a better way to compute the inverse of A.

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- Example: Find the inverse of the matrix.

$$A=\begin{bmatrix}2 & 1 & 3\\ 1 & -1 & 1\\ 1 & 4 & -2\end{bmatrix}$$

Solution: The nine cofactors are

$$C_{11}=+\begin{vmatrix}-1 & 1\\ 4 & -2\end{vmatrix}=-2, \quad C_{12}=-\begin{vmatrix}1 & 1\\ 1 & -2\end{vmatrix}=3, \quad C_{13}=+\begin{vmatrix}1 & -1\\ 1 & 4\end{vmatrix}=5$$

$$C_{21}=-\begin{vmatrix}1 & 3\\ 4 & -2\end{vmatrix}=14, \quad C_{22}=+\begin{vmatrix}2 & 3\\ 1 & -2\end{vmatrix}=-7, \quad C_{23}=-\begin{vmatrix}2 & 1\\ 1 & 4\end{vmatrix}=-7$$

$$C_{31}=+\begin{vmatrix}1 & 3\\ -1 & 1\end{vmatrix}=4, \quad C_{32}=-\begin{vmatrix}2 & 3\\ 1 & 1\end{vmatrix}=1, \quad C_{33}=+\begin{vmatrix}2 & 1\\ 1 & -1\end{vmatrix}=-3$$

- Thus

$$\operatorname{adj} A = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix}$$

Check the calculations and produces det A:

$$(\operatorname{adj} A) \cdot A = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix} = \begin{bmatrix} 14 & 0 & 0 \\ 0 & 14 & 0 \\ 0 & 0 & 14 \end{bmatrix} = 14I$$

Since (adj A)A = 14I, Theorem 8 shows that det$A$ = 14 and

$$A^{-1} = \frac{1}{14} \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix} = \begin{bmatrix} -1/7 & 1 & 2/7 \\ 3/14 & -1/2 & 1/14 \\ 5/14 & -1/2 & -3/14 \end{bmatrix}$$

§3.3 Cramer's Rule, Volume, and Linear Transformations

Determinants as Area or Volume（几何意义）

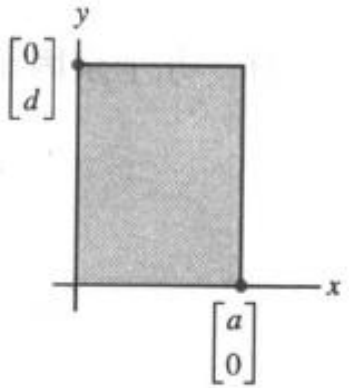
Theorem 9

If A is a 2×2 matrix, the area of the parallelogram (平行四边形) determined by the columns of A is |det A|. If A is a 3×3 matrix, the volumn of the parallelepiped(平行六面体)determined by the columns of A is |det A|.

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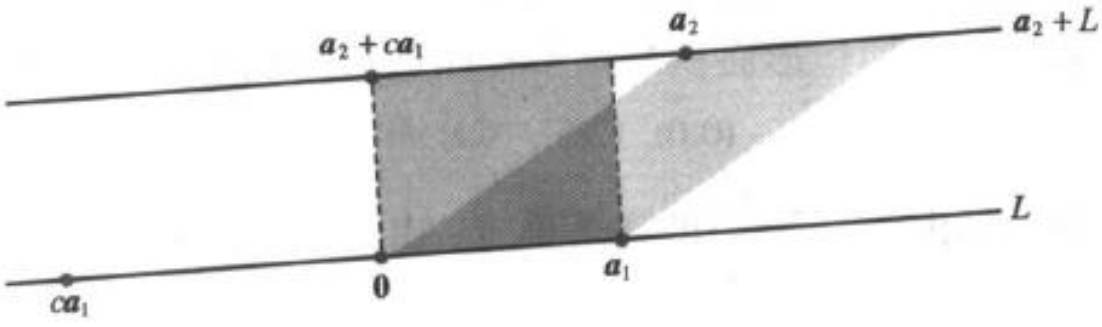
Proof : The theorem is true for any 2×2 diagonal matrix:

$$\left|\det\begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}\right| = |ad| = \{\text{矩阵的面积}\}$$



- For other 2×2 matrix A = [$\mathbf{a}_1$ $\mathbf{a}_2$], It will suffice to show that A can be transformed into a diagonal matrix.

Let $\mathbf{a}_1$ and $\mathbf{a}_2$ be nonzero vectors. Then for any scalar *c*, the area of the parallelogram determined by $\mathbf{a}_1$ and $\mathbf{a}_2$ equals the area of the parallelogram determined by $\mathbf{a}_1$ and $\mathbf{a}_2 + c\mathbf{a}_1$.



- For 3×3 matrix A = $[\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3]$,

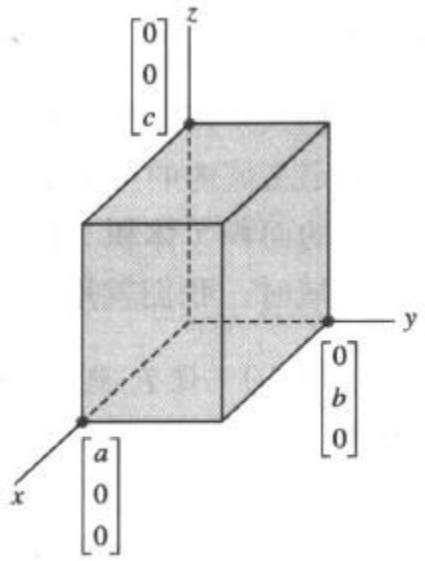
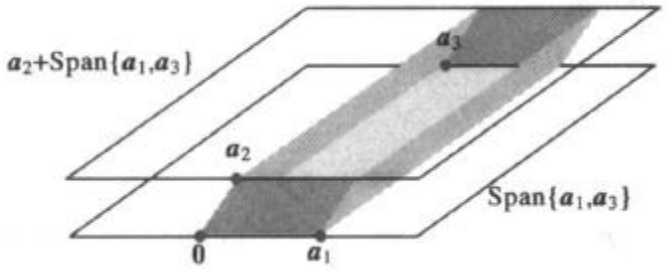
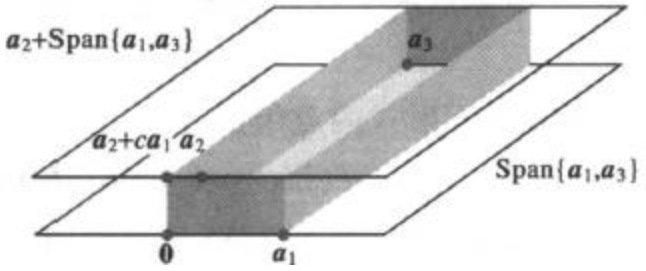


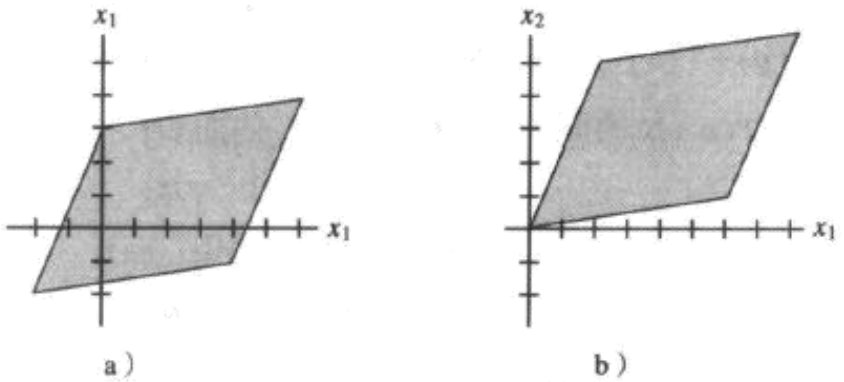
图 3-6　体积=|a b c|

Two parallelograms of equal area





- Example:Calculate the area of the parallelogram determined by the points $(-2,2),(0,3),(4,-1)$和$(6,4)$



- Solution : First translate the parallelogram to one having the origin as a vertex. The new parallelogram has the same area, its vertices are $(0,0),\ (2,5),\ (6,1)$ 和 $(8,6)$

It is determined by the columns of $A=\begin{bmatrix}2 & 6\\ 5 & 1\end{bmatrix}$

Since |det A|= |-28|, the area of the parallelogram is 28.

§3.3 Cramer's Rule, Volume, and Linear Transformations（了解）

Linear Transformations

Theorem 10

Let $T:R^2 \to R^2$ be the linear transformation determined by a 2×2 matrix A. If S is a parallelogram in R^2, then

$$\{\text{area of } T(S)\} = |\det A| \cdot \{\text{area of } S\}$$

If T is determined by a 3×3 matrix A, and if S is a parallelepiped in R^3, then

$$\{\text{volume of } T(S)\} = |\det A| \cdot \{\text{volume of } S\}$$

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- Proof : Consider the 2×2 case, with A = [$\mathbf{a}_1$ $\mathbf{a}_2$]. A parallelogram at the origin in R^2 determined by vectors b_1 and b_2 has the form

$$S = \{s_1\boldsymbol{b}_1 + s_2\boldsymbol{b}_2 : 0 \leqslant s_1 \leqslant 1,\ 0 \leqslant s_2 \leqslant 1\}$$

The image of S under T consists of points of the form

$$T(s_1\boldsymbol{b}_1 + s_2\boldsymbol{b}_2) = s_1T(\boldsymbol{b}_1) + s_2T(\boldsymbol{b}_2) = s_1A\boldsymbol{b}_1 + s_2A\boldsymbol{b}_2$$

So T(S) is the parallelogram determined by the columns of the matrix [Ab_1 Ab_2], written as AB. By Theorem 9,

$$\{T(S)\text{的面积}\} = |\det AB| = |\det A| \cdot |\det B|$$
$$= |\det A| \cdot \{S\text{的面积}\}$$

- The conclusions of Theorem 10 hold whenever S is a region in R^2 with finite area or a region in R^3 with finite volumn.

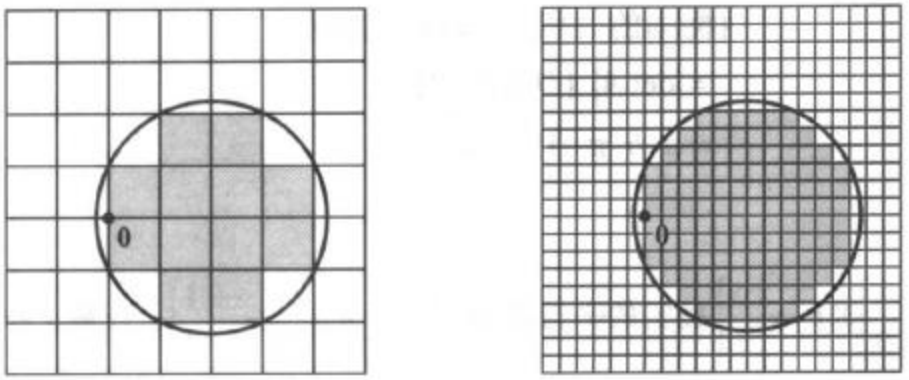


图 3-9　由正方形近似一个平面区域. 近似的程度随格子变细小而改进

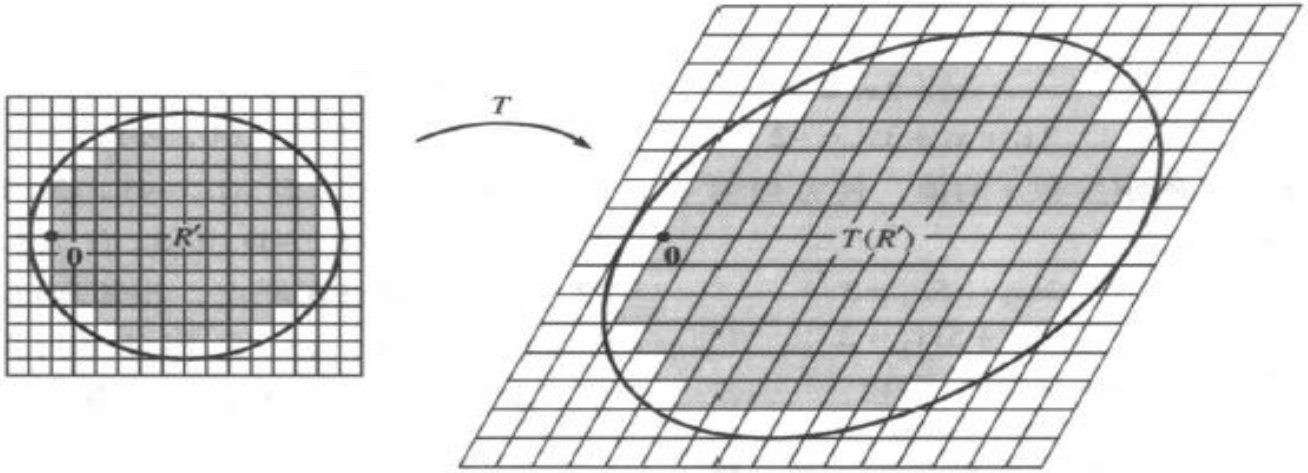


图 3-10　$T(R)$ 由平行四边形的并集近似

Assignment

- Ex3.3: 7, 11，**21**

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